

Beyond Metacognitive Accuracy: A Computational Account of Confidence Updating Dynamics in Schizophrenia

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Abstract

Abnormal confidence judgements that do not reflect reality may influence pathological decision making typically observed in schizophrenia spectrum disorders. These confidence abnormalities underpin adaptive behaviour, daily functioning and treatment response in schizophrenia spectrum disorders (SSD), and are therefore important to understand. While confidence abnormalities are found in positive association with symptom severity, recent work comparing confidence dynamics of control groups and individuals with SSD have revealed no significant differences at the group level. These findings contradict each other, therefore we decided to investigate at a trial level, the accumulation of evidence post-decision, often characterised as confidence formation.

This study aims to investigate differences in the confidence accumulation during a visual discrimination task in healthy controls and individuals with schizophrenia spectrum disorder at the trial level. We apply a two stage dynamic signal detection framework to characterise confidence accumulation rates post -decision in individuals with SSD and healthy controls (HC). This study quantifies and compares confidence accumulation rate differences using behavioural parameters including response time and choice with an evidence accumulation model. The same dataset previously reported to show null group-level differences was used here to examine whether trial-level dynamics diverge even where group-level metacognitive efficiency does not. Since this dataset used continuous confidence ratings, the two stage dynamic model was customised to continuous (as opposed to the original discrete) confidence values.

Results show both groups performed similarly as far as reporting performance accuracy was concerned (i.e. metacognitive efficiency is intact). However, SSD individuals accumulated confidence at nearly half the rate of healthy controls after perceptual decision commitment.

These findings extend existing models by pinpointing altered confidence dynamics specifically in the post-decisional phase.

Keywords: Schizophrenia, Confidence, Metacognition, Two Stage Dynamic Signal Detection Model, HDDM.

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List of abbreviations

2DSD	Two-Stage Dynamic Signal Detection
CT-R	Recovery-Oriented Cognitive Therapy
DDM	Drift Diffusion Model
DIC	Deviance Information Criterion
ESS	Effective Sample Size(s)
ETI	Equal-Tailed Interval
HC	Healthy Control(s)
HDDM	Hierarchical Drift Diffusion Model
IQ	Intelligence Quotient
MCMC	Markov Chain Monte Carlo
NUTS	No-U-Turn Sampler
PyMC	Python-based probabilistic programming package
RDK	Random Dot Kinetogram
RT	Reaction Time

SDT	Signal Detection Theory
SSD	Schizophrenia Spectrum Disorder(s)
VAS	Visual Analogue Scale

1. Introduction

People often know when they are right, versus when they are not. Evolutionarily speaking, being right (for example, about the poisonous nature of a berry) could mean the difference between life and death for humans. So , Man constantly monitors his own accuracy while making decisions in real time, a phenomenon referred to as metacognition. Metacognition is the cognitive ability to control, monitor and modulate cognitive processes (Martiadis et al, 2023) thus guiding and orienting behaviour; such as the confidence in accuracy of one's judgement. Confidence, in particular, has been widely acknowledged as a behavioural expression of metacognition, because it reflects an internal estimate of the reliability of a decision (Fleming and Lau, 2014). It is no surprise therefore, that systematically inaccurate confidence judgments could contribute to pathological decision making such as in the case of

schizophrenia spectrum disorders (Hoven et al, 2019). Schizophrenia spectrum disorders specifically, are associated with overconfidence in inaccurate judgements and having fixed false beliefs (delusions) about the world around them (Warman et al, 2008). Since confidence guides future decisions and behaviours in the absence of external feedback, understanding how confidence is generated and updated in schizophrenia spectrum disorders might provide insight into the mechanisms underlying the pathology. One increasingly popular way of doing this is to mathematically model the decision process and the accumulation of evidence after the perceptual decision i.e. the confidence associated with the decision being made.

Over the past several decades, drift diffusion models (DDM) have emerged as a dominant framework describing perceptual decision making (Ratcliff, 1978). These models explain choice and reaction time by assuming that noisy sensory evidence is accumulated until a decision threshold is reached. Although these models have served to advance our understanding of decision formulation, they do not explain how those decisions are evaluated after they are made (Pleskac and Busemeyer, 2010). This is why since the drift diffusion model, several computational models have approximated confidence formation underlying decision making. Signal detection theory (SDT) conceptualises confidence as the strength of an internal decision variable relative to one or more confidence criteria, allowing confidence judgements to be modelled independently of decision accuracy (Green and Swets, 1966, Maniscalco and Lau, 2012). The drift diffusion model discussed earlier has extended this by correlating confidence with the rate at which evidence is accumulated after the point of decision commitment (Kiani and Shadlen, 2009). The Two Stage Dynamic Signal Detection model proposed by Pleskac and Busemeyer (2010), builds on this, it suggests that evidence accumulation continues following the decision, and the post-decisional evidence correlates with confidence. This offers a tangible way in which confidence accumulation can be tracked

during the decision process, and approximate the metacognitive efficiency (accurate confidence judgement) in individuals characterised with low metacognitive efficiency.

Studies involving individuals with schizophrenia spectrum disorders (SSD) have revealed abnormalities in confidence reporting, maladaptive beliefs, and impaired introspective accuracy (Kircher et al, 2007; Bhatt et al, 2010; Balzan, 2016). However, Faivre et al (2023) , among other studies, found no evidence of a specific metacognitive deficit across groups in a visual discrimination task after carefully controlling for task accuracy. The original study reported that confidence judgements and metacognitive performance were largely preserved at the group level in individuals with schizophrenia, suggesting that previously reported deficits might have been confounded by differences in cognitive performance rather than metacognitive dysfunction (Faivre et al, 2023). Individuals with SSD tend to perform worse in cognitive tasks as compared to HC (Gurrera et al, 2014). These null findings raise an important question regarding the computational mechanisms underlying confidence: the absence of group differences in overall metacognitive efficiency does not imply that confidence is generated or utilised identically across HC and SSD groups. The present thesis approaches the same dataset to determine whether the behavioural equivalence reported by Faivre et al (2023) at the group level conceals differences in evidence accumulation, confidence formation, updating or influence of confidence on following decisions at the trial level.

The present thesis investigates whether confidence acts as a metacognitive feedback signal that dynamically updates subsequent decision making, and whether this mechanism differs between individuals with SSD and HC across trials. The proposed framework conceptualises confidence as a dynamic variable that is generated post decision formation, updated through experience and used to inform future decisions. This extends the work of Faivre et al (2021),

who examined metacognitive efficiency in individuals with SSD and healthy controls at the group level using drift diffusion modelling. The present study investigates whether there is a difference in how this confidence is generated and used in future decisions across the two groups at the trial level. The present analysis therefore tests confidence as a dynamic feedback signal, not the outcome of a decision.

This question is addressed using a two stage computational framework, with each stage targeting a different component of the confidence generating and updating process. Stage one applies a Hierarchical Drift Diffusion Model (HDDM; Wiecki et al , 2013) to examine whether confidence reported on one trial predicts changes in latent decision parameters (drift rate, boundary separation, etc) on the following trial, and whether these relationships differ between SSD and HC. Desender et al. (2019) demonstrated that confidence reported on one trial influenced the decision criterion adopted on the subsequent trial, suggesting that confidence can be used to regulate future decision-making. The hierarchical formulation estimates both the individual trial level effects and the group level effects, it allows information to be shared across participants while retaining participant-level variation, producing more stable estimates (Wiecki et al, 2013). The model therefore analyses whether previous confidence changes the subsequent decision threshold or efficiency of evidence accumulation.

Stage two addresses the question: how is confidence generated during the decision process?

The Two Stage Dynamic Signal detection (2DSD) model proposed by Pleskac and Bussemeyer (2010) extends the evidence accumulation account by proposing that evidence continues to accumulate after the decision process, and the rate of this this post-decision evidence accumulation corresponds to the confidence associated with that decision. Thus, where Stage one treats confidence from the previous trial as input to the next decision, stage

two models confidence as an output of the current decision process. This allows the two stages to address different directions of the proposed feedback relationship: stage one tests whether the confidence subsequently influencing decision formation differs in SSD and HC, and stage two tests whether the confidence generation process after a decision is different in the two groups. Together, these analyses provide a computational account of confidence both as a product of decision formation and an internal signal for future decisions, while extending the analysis of Faivre et al (2023) dataset to examine the dynamic role of confidence in SSD.

The present framework examines confidence across two interrelated processes : generation during decision formation and its influence on future decision making. The aim of this work is therefore to develop a mechanistic account of confidence as both the outcome of decision formation and a potential source of information for subsequent behavioral adaptation, and to examine these processes in SSD. In doing so, the thesis makes four principal contributions:

- Mechanistic account of confidence in SSD : examines whether differences in confidence arise from evidence accumulation, response caution, confidence formation or how it is utilised in the next decision.
- Extension of 2DSD to continuous confidence: adapts the 2DSD model to continuous confidence ratings, allowing post-decisional evidence accumulation to be examined without reducing confidence to discrete categories.
- Confidence as a dynamic signal: tests whether previous confidence influences subsequent decision-making and whether this differs between SSD and HC.

2. Methods

To address the research question of whether metacognitive impairments in SSD arise from altered post-decisional evidence accumulation, the two stage computational modelling approach described above was employed. The analyses were designed to separate processes involved in the initial perceptual decision from those involved in subsequent confidence formation. First, hierarchical drift diffusion models (HDDMs) were fitted to participant's choices and reaction times to characterise latent decision-making parameters and determine whether any group differences could be explained by first order decision processes. Building on these findings, the 2DSD model was then implemented to quantify post-decisional evidence and examine whether the dynamics of confidence formation differed between healthy controls and individuals with SSD (Pleskac and Busemeyer, 2010). The following sections describe the dataset , preprocessing procedures, and implementation of each modelling stage.

2.1. Dataset:

The present study involved the secondary analysis of an existing, fully anonymised behavioural dataset obtained from a publicly available open-access repository, OpenNeuro. The study involved no primary human participant research, as no identifiable personal information was available to the researcher.

The original study (Faivre et al, 2021) recruited 23 healthy control (HC) participants from the general population and 20 participants with schizophrenia spectrum disorder (SSD) from community mental health centres and outpatient clinics in the Versailles area. Following data collection, two healthy control participants were excluded from the original analysis, one due to convergence failure during the adaptive staircase procedure and one due to an estimated

intelligence quotient (IQ) below 70. One participant with SSD was also excluded because of incomplete trial data, resulting in a final sample of 21 HC and 19 participants with SSD.

Participants in both groups met the original study's eligibility criteria (Faivre et al, 2021). Exclusion criteria included moderate-to-severe substance use disorder within the preceding 12 months, current or previous untreated medical or neurological illness, an estimated IQ below 70 based on three subtests of the Wechsler Adult Intelligence Scale, and age greater than 60 years (Wechsler, 2008). All participants were right-handed, had normal or corrected-to-normal vision, normal hearing, and were naïve to the purpose of the study. Written informed consent was obtained from all participants prior to participation. Study information was provided orally and adapted to the level of comprehension of participants with schizophrenia spectrum disorder, with written information also supplied. The original study received ethical approval from the Sud Méditerranée II Research Ethics Committee (Faivre et al , 2021).

2.2 Experimental design

In each trial, participants were presented with a random dot kinetogram (RDK) and were asked to indicate which direction they thought a majority of the dots were moving. The two directions indicated were either right or left. Once this was completed, they were asked to indicate how confident they were in their decision on a scale of 0-100 by dragging a cursor along a visual analog scale (VAS) (Faivre, et al, 2021).

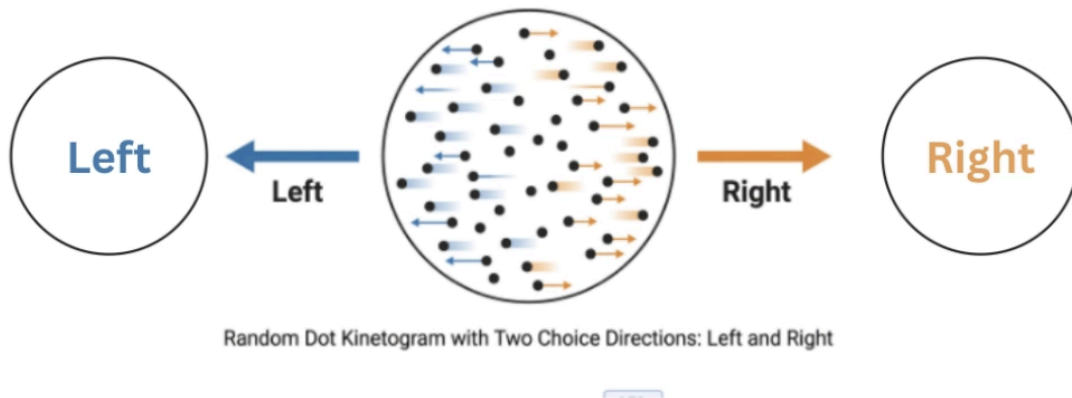


Fig 1. Random dot kinetogram

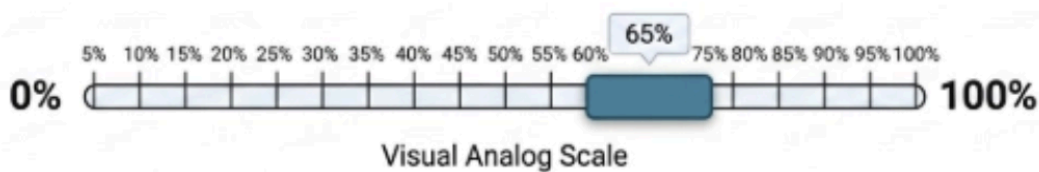


Fig 2. VAS scale

2.2.a Staircase design

The staircase design as implemented by the original study manipulates the standard deviation of the dot directions, the stimulus dispersion. It controls the angles and dispersion of the dot movement in the stimulus. At the start of the experiment the staircase is initialised such that initial dispersion is the participants' threshold estimated from the mean of the last six training sessions. The staircase operates on a fixed step size of 2 degrees, and updates the stimulus difficulty on every trial using a 1-up/2-down update rule. According to the 1-up/2-down rule, two consecutive correct responses were required before dispersion is increased (making the stimulus harder), whereas a single incorrect response immediately decreases the dispersion (making the next stimulus easier). The staircase keeps adjusting the angular dispersion of the

motion stimulus so that participants converge at roughly 71% accuracy. This adaptive procedure as designed by Faivre et al (2021) ensures that HC and SSD complete tasks at comparable levels of subjective accuracy. Confidence was reported after every trial by dragging a cursor along a VAS scale with marks at 5% increments from 0-100%. It is important to note here, that the staircase algorithm tracked only accuracy, and not the reported confidence of the participant after every trial.

This is relevant when modelling reaction times (rt) and choice, as we will do, because differences in these measures may be otherwise difficult to distinguish from differences in the stimulus difficulty itself. Participants did not receive any feedback regarding their performance throughout the data collection. The staircase algorithm used the accuracy of their response to determine the difficulty of the next trial. Therefore, while the participants were not aware of their performance accuracy explicitly, their trials were influenced by previous responses. This study identifies confidence as the signal elicited immediately following the perceptual decision and therefore reflected the participant's subjective evaluation of their decision on the previous trial without explicit feedback (Desender, 2021). The resulting confidence measure can therefore be considered in relation to both the evidence available during the initial decision and the information accumulated following the decision.

The trial level structure of the dataset was retained throughout preprocessing and modelling. Each observation for a participant's trial included a participant identifier, group (HC/SSD), trial number, binary choice (left/right), reaction time, confidence rating, accuracy and the trial specific stimulus and staircase dispersion information. The reaction times included time of stimulus onset, time at which the participant started moving the mouse towards direction of choice, and time at which they clicked on the choice. This structure allowed each choice and reaction time to be modelled across two stages of the decision process while maintaining the

temporal structure of the decision process and subsequent confidence report. In addition, variables derived from the preceding trial, including previous confidence, were retained for analyses examining the influence of confidence on previous trials on the next one.

2.3 Preprocessing

Preprocessing as conducted at the trial level to retain the structure of the data required to model decision and confidence formation across the two stages. Movement onset was defined as the first point at which mouse velocity exceeded 20% of the trial-wise peak velocity, following the procedure used in Faivre et al (2021). Decision reaction time (rt) was measured from stimulus onset (as the stimulus appeared on the screen), to the movement onset (as the mouse starts moving). Post decisional or reach duration was defined as the duration from movement onset to the click (response). This distinction was made because movement represents the participant's perceptual decision, whereas the reach period provides the measured interval between decision and confidence accumulation.

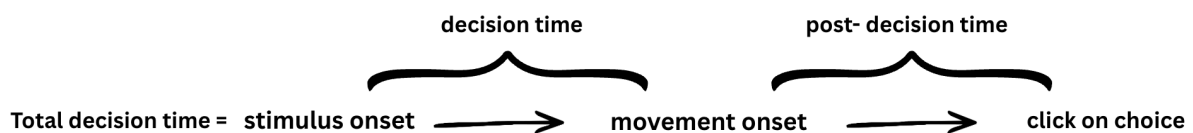


Fig. 3 Decision reaction time

Confidence ratings were converted from the original 0-100 scale to 0-1. Ratings were not mean-centered or standardised because the analysis is designed to examine both confidence level and variability; removing individual differences in these measures during preprocessing would remove information relevant to the research question. Participant-level differences were accounted for within the hierarchical models. Raw group means and standard deviation were calculated separately as descriptive measures and were not used to determine exclusions

or model specifications. Trials with missing confidence responses were excluded because they could not contribute to the confidence analysis. Trials with implausible reaction times or reach durations were also removed following inspection of individual participant distributions. This allowed clearly invalid observations to be identified while avoiding a single overly restrictive cut-off across participants. The final dataset therefore preserved trial-level information relevant to both initial decision-making and subsequent confidence formation while excluding observations that could not provide reliable model estimates.

At the end of the preprocessing stage, approximately 300 trials for each participant with 21 in HC and 19 in SSD groups were retained.

2.4 Stage 1: HDDM

Differences in observed accuracy and reaction times alone are insufficient to determine the source of any group differences in the decision process. Comparable accuracy between groups (as has been established in this paradigm) could mask differences in how efficiently evidence is accumulated or in the amount of evidence required before committing to a choice. It is therefore not possible to infer where in the decision process group differences arise using accuracy and reaction time data alone.

Modelling the initial decision provides a way of locating where in the decision-evaluation cycle group differences occur. Hierarchical Drift Diffusion models (HDDM) was used to decompose observed behaviour into latent cognitive components. Drift diffusion models extend signal detection theory by proposing a mechanistic relationship between decision outcomes and reaction times (Ratcliff and McKoon, 2008). They assume that once the evidence accumulated towards a specific decision crosses a pre-defined threshold, a decision is made. To examine this, a hierarchical drift diffusion model (HDDM) was implemented in a

Docker container with the HDDM python package (Wiecki et al, 2013). The HDDM package uses Bayesian hierarchical estimation, which allows individual participant level parameters to be modelled as they are drawn from group level distributions. This hierarchical structure produces partial pooling, whereby parameter estimates from participants with relatively few observations are regularised towards the group mean while participants with substantial data retain greater influence over their individual estimates. Compared with fitting each participant independently, hierarchical estimation produces more stable and statistically efficient parameter estimates, particularly in clinical datasets where trial numbers are moderate and behavioural variability is high. HDDM estimates three latent decision parameters : drift rate (v), boundary separation (a) and non-decision time (t).

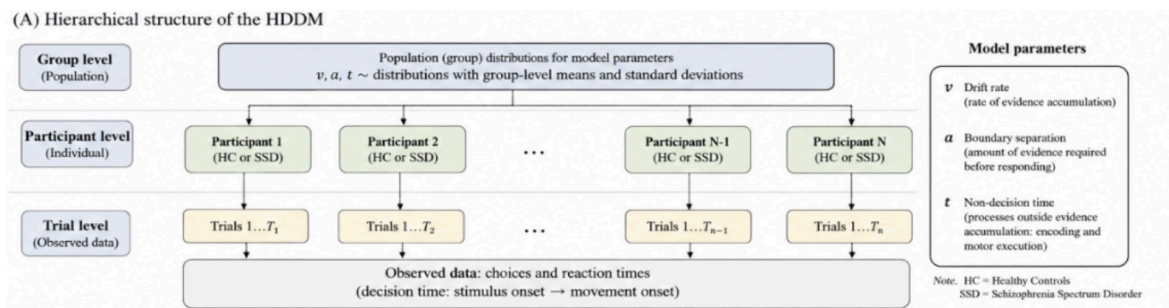


Fig 4. HDDM structure for Stage 1

Drift rate (v) is the average rate at which evidence is accumulated towards a response, while boundary separation is the amount of evidence required before a decision is made . The non-decision time refers to the amount of time during which evidence is not accumulated, i.e. in this study, how long the participant takes to perceive the stimulus post-onset (Ratcliff and McKoon, 2008). Together, these parameters allow differences in reaction times and choices to be decomposed into differences in rate of evidence accumulation, decision threshold and processes occurring outside the decision process. The HDDM thus estimates the latent evidence accumulation process that produces the initial choice. In this stage, decision time was identified as the time between stimulus onset and movement onset. Confidence represents

an internal evaluation of the reliability of a previous decision and therefore provides a plausible signal for adapting subsequent decision policy. After a low confidence decision, participants may compensate by adopting a more conservative response criterion, requiring more evidence before committing to the next choice, whereas high confidence may promote a less cautious strategy. So, previous-trial confidence was hypothesised to influence boundary separation.

Previous confidence was calculated for each trial as the confidence reported on the immediately preceding trial by the same participant. The first trial of every participant (which had no previous confidence value) was not included in this stage. To examine whether the effect of previous confidence differed according to the participant's level of prior confidence, previous confidence values were divided into two within-participant groups using the participant-specific median : low previous confidence and high previous confidence. This within participant split ensured that comparison reflected changes in confidence relative to each participant's own reporting range and was not influenced by differences in overall confidence scores between participants. This discretisation was also necessary because the `depends on` function in the HDDM package requires predictors to be specific in discrete bins, not the continuous confidence scale available in the dataset (Wiecki et al, 2013). At this level of analysis however, exact confidence values were not strictly required, and so this trade-off was made. Ties at the median were assigned to the high category, and the resulting variable was used to test whether the relationship between previous confidence and subsequent decision process differed between HC and SSD participants.

2.4 a. Model specifications:

Four hierarchical models of increasing complexity were fitted to determine which latent processes were influenced by group and previous confidence.

Model	v	a
Model 1	$v \sim \text{group}$	constant
Model 2	$v \sim \text{group}$	$a \sim \text{previous confidence}$
Model 3	$v \sim \text{group}$	$a \sim \text{group} \times \text{previous confidence}$
Model 4	$v \sim \text{group} \times \text{previous confidence}$	$a \sim \text{group} \times \text{previous confidence}$

Table 1. Stage 1 model specifications

Model 4 was the most flexible model, allowing both evidence accumulation and response caution to differ across every combination of diagnostic group and previous confidence. All model were fitted with non-decision time (t) held constant across conditions. Following model comparison, a separate control model additionally allowed $t \sim \text{group}$ to evaluate if any observed group differences in sensory encoding or motor preparation rather than evidence accumulation. This model did not substantially improve model fit and was not retained. Models were compared using the deviance information criterion (DIC), with lower values indicating a better trade off between goodness-of-fit and model complexity.

For the winning model, posterior distributions were examined for the estimated ‘v’ and ‘a’ parameters in each of the conditions. Posterior means and 95% credible intervals were calculated for each parameter, together with posterior contrasts between all the conditions. These contrasts allowed the analysis to determine whether the groups differed in evidence accumulation or decision threshold and whether the effect of previous confidence on these

processes differed between groups. These estimates provide a more direct assessment of the latent decision processes than the observed reaction times and choices alone.

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The Stage 1 analysis establishes whether differences between HC and SSD are already present during the initial perceptual decision and whether the preceding confidence influences the next decision process. However, Stage 1 leaves questions about how this confidence is generated unanswered. The HDDM uses observed parameters, i.e. choices and reaction times to characterise the initial decision but does not represent the evidence that may continue to accumulate after the decision boundary has been reached. It cannot determine

whether differences in confidence arise from differences in post-decisional evidence accumulation, nor can it describe how this evidence is reported as a continuous confidence judgement. To address these remaining questions, the second stage implements a 2DSD model which represents the evidence accumulation following the initial perceptual decision and models its relationship with the confidence report.

2.5. Stage 2: 2DSD model

To characterise the post-decisional processes underlying confidence judgements, confidence ratings were analysed using the 2DSD model (Pleskac and Busemeyer, 2010). conventional DDM can account for the formation of the initial choice, but does not identify the computational mechanism for confidence accumulation. The 2DSD framework extends the drift diffusion model by proposing that evidence accumulation does not terminate once a decision boundary has been reached. Instead, evidence continues to accumulate during a second, post-decisional stage, with the resulting evidence state forming the basis of subjective confidence. This framework provides a mechanistic account of confidence as an extension of the same evidence accumulation process that generated the initial perceptual decision.

Unlike the original 2DSD model, which was developed for discrete confidence judgements and estimates post-decisional accumulation time was a free parameter (Pleskac and Busemeyer, 2010), the present implementation was adapted to accommodate continuous confidence ratings and directly observed post-decision accumulation intervals. Consistent with the experimental design of Faivre et al (2021), post -decisional accumulation time (τ) was defined as the interval between movement onset and the confidence response. Because this interval was recorded on every trial, (τ) was treated as an observed quantity rather than an unknown parameter requiring estimation.

A second modification concerned the duration of post-decisional evidence accumulation. In the original 2DSD model, post-decisional accumulation time is treated as an unknown quantity that must be estimated from the data. In the present thesis, however, the interval between movement onset and the click response was recorded every trial. Following the design of Faivre et al (2021), this interval was defined as the post-decisional accumulation time and entered directly into the model as an observed variable. Conditioning the model on measured accumulation times reduced the number of free parameters while ensuring that predicted confidence reflected the actual duration over which post-decisional evidence could accumulate on each trial.

$$L_{tc,i} \sim N(\mu_{L,i}, \tau_i) \dots\dots\dots(Equation 1)$$

where $\mu_{L,i}$ is the expected value (mean) of the latent evidence distribution and τ_i is the observed post decision interval between the perceptual response and the subsequent confidence report. The variance of the latent evidence therefore increases linearly with post-decision time, consistent with equation 13 of the original 2DSD model,

$$var(L_{tc,i}) = \sigma^2 \tau_i \dots\dots\dots(Equation 2)$$

Following the standard identifiability constraint used in diffusion models, the diffusion noise scaling parameter was fixed at $\sigma = 1$, reducing the variance to

$$var(L_{tc,i}) = \tau_i \dots\dots\dots(Equation 3)$$

The standard deviation of the latent evidence distribution is $\sqrt{\tau_i}$, meaning that uncertainty surrounding the accumulated evidence increases as post decisional accumulation continues.

The expected value of the latent evidence distribution followed Equation 12 of Pleskac and Busemeyer (2010),

$$\mu_{L,i} = \theta + \delta_{gi}\tau_i \dots\dots\dots(Equation 4)$$

where θ represents the residual decision evidence immediately after the initial perceptual choice and before post-decisional accumulation begins. This parameter therefore determines the starting point of the second-stage accumulation process and was constrained to positive values using a Half-Normal prior,

$$\theta \sim \text{HalfNormal} \dots\dots\dots(Equation 5)$$

The parameter δ_{gi} represents the post-decisional drift rate, that is, the average rate at which evidence continues to accumulate following the initial decision. Unlike the original 2DSD model, a separate drift parameter was estimated for each group -

$$\delta = \begin{bmatrix} \delta_{\text{HC}} \\ \delta_{\text{SSD}} \end{bmatrix} \dots\dots\dots(Equation 6)$$

with each parameter assigned a weakly informative Normal prior,

$$\delta_g \sim N(0,1) \dots\dots\dots(Equation 7)$$

Estimating separate post decisional drift rates enables direct testing of the hypothesis that schizophrenia spectrum disorder is associated with altered post-decisional evidence accumulation during confidence formation.

Although the latent evidence distribution could be sampled directly from the Normal distribution above, a non-centered parameterisation was employed to improve Hamiltonian Monte Carlo sampling efficiency and reduce posterior correlations between parameters. Specifically,

$$z_i \sim N(0, 1) \dots\dots\dots(Equation 8)$$

and the latent evidence defined as

$$L_{tc,i} = \mu_{L,i} + \sqrt{\tau_i} z_i \dots\dots\dots (Equation 9)$$

This formulation is mathematically identical to sampling

$$L_{tc,i} \sim N(\mu_{L,i}, \tau_i) \dots\dots\dots (Equation 10)$$

but generally results in faster convergence, improved effective sample sizes and fewer divergent transitions in hierarchical Bayesian models.

The original 2DSD model predicts the probability of selecting one of the several discrete confidence categories according to

$$P(C = j) = P(c_j < L_{tc} < c_{j+1}) \dots\dots\dots (Equation 11)$$

where c_j and c_{j+1} , represent adjacent confidence criteria separating ordinal confidence responses. However, participants in the present study reported confidence on a continuous scale between 0 and 1 rather than selecting discrete confidence categories. The threshold-based confidence model was replaced with a continuous mapping between latent evidence and confidence.

Expected confidence was obtained by transforming the latent evidence using the cumulative standard normal distribution,

$$\mu_{c,i} = \phi(L_{tc,i}) \dots\dots\dots (Equation 12)$$

where $\phi(\cdot)$ denotes the cumulative distribution function of the standard Normal distribution (implemented in PyMC using the inverse probit transformation). This nonlinear transformation maps the unbounded latent evidence variable,

$$L_{tc} \in (-\infty, \infty) \dots\dots\dots (Equation 13)$$

onto the bounded confidence interval

$$\mu_c \in (0, 1) \dots\dots\dots (Equation 14)$$

thereby matching the scale of the observed confidence ratings while preserving the assumption that greater accumulated evidence corresponds to higher subjective confidence.

Observed confidence ratings, denoted c_i , were assumed to represent noisy measurements of this latent confidence state were modelled as

$$c_i \sim N(\mu_{c,i}, \sigma_c) \dots\dots\dots (Equation 15)$$

where σ_c captures residual variability in confidence reports that is not explained by the post-decisional accumulation process. This observation noise parameter was assigned a weakly informative HalfNormal prior,

$$\sigma_c \sim \text{HalfNormal}(0.5) \dots\dots\dots (Equation 16)$$

Prior to model fitting, confidence ratings were clipped to the interval $[10^{-5}, 1 - 10^{-5}]$ to avoid numerical instabilities associated with applying the cumulative normal transformation to observations lying exactly on the boundaries of the confidence scale.

Posterior distributions were estimated using the No-U-Turn Sampler (NUTS) implemented in PyMC. Four independent Markov chains were sampled, each comprising of 2000 tuning iterations followed by 5,000 posterior samples. The target acceptance probability was increased to 0.95 to improve sampling stability for the hierarchical latent variable model.

Model convergence was assessed using trace plots, the Gelman-Rubin statistic ($\hat{R}$), and effective sample sizes (ESS).

The principle parameter of interest was the difference in post-decisional drift rate between groups,

$$\Delta_{\delta} = \delta_{SSD} - \delta_{HC} \dots\dots\dots(Equation 17)$$

for which posterior means, 95% credible intervals, and the posterior probability,

$$P(\delta_{SSD} > \delta_{HC}) \dots\dots\dots(Equation 18)$$

were calculated. This posterior contrast directly quantified the probability that participants with schizophrenia accumulated post-decisional evidence at a different rate from healthy controls during confidence formation.

In addition to the group differences in the post-decisional drift rate, the posterior distributions were calculated for all parameters governing confidence formation, including the θ , δ_{HC} , and δ_{SSD} , and the σ_{noise} . These posterior distributions were used to estimate the most likely value of each parameter while retaining uncertainty around the estimate. Convergence was assessed using trace plots, $\hat{R}$, and effective sample size (ESS). Parameter estimates were considered reliable when the chains mixed adequately, $\hat{R}$ was close to 1, and ESS were sufficient to provide stable estimates of the posterior distributions.

Posterior uncertainty was evaluated using posterior means, and 95% credible intervals. Group differences were assessed directly from the posterior distributions and the posterior probability of $P(\delta_{SSD} > \delta_{HC})$ was also calculated, providing a direct estimate of the probability that post-decisional evidence accumulation was greater in SSD than HC. Posterior distributions for HC and SSD were compared for the initial evidence state, post-decisional drift rate, and confidence observation noise where appropriate. These comparisons allowed the analysis to distinguish differences in the starting level of post-decisional evidence from

differences in the rate at which evidence accumulated and from differences in the variability of reported confidence. The resulting posterior estimates were therefore interpreted at the level of the confidence-generation process rather than solely at the level of observed confidence ratings.

The output of Stage 2 is the posterior distributions for how evidence was accumulated post decision for HC and SSD groups. A lower post-decisional drift rate in SSD would indicate that evidence supporting the initial decision accumulated more slowly during the period in which confidence was formed. Differences in θ would instead indicate differences in the evidence state immediately following the initial decision, while differences in σ_c would indicate differences in the extent to which reported confidence varied around the model-predicted confidence state. Together, these estimates provide a process-level account of confidence formation.

3.Results

3.1. Stage 1 : HDDM

The first analysis tested whether previous trial confidence influenced subsequent decision formation, and whether this relationship differed between HC and SSD participants. The analysis examined whether group and previous confidence affected the rate of evidence accumulation (v), and the amount of evidence required before making a decision (a). To address these questions, four hierarchical drift diffusion models were compared using the Deviance Information Criterion (DIC).

Model	DIC	Δ DIC
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Model 1 45549.97 1552.88

Model 2 45416.16 19.07

Model 3 45415.42 18.33

Model 4 45397.09 0.00

Table 2. Model DICs

The models progressively allowed ‘v’ and ‘a’ to vary to group and previous confidence conditions. Model 4, in which both parameters varied according to the interaction between groups and previous confidence, provided the best fit to data. For DIC values, refer to Table (). The lowest DIC supported the model in which both evidence accumulation and decision threshold depended on the combination of the diagnostic group and previous confidence.

3.1.a. Drift rate :

The drift rate estimates were examined to determine whether the rate of evidence accumulation on the current trial differed between groups and according to confidence on the preceding trial.

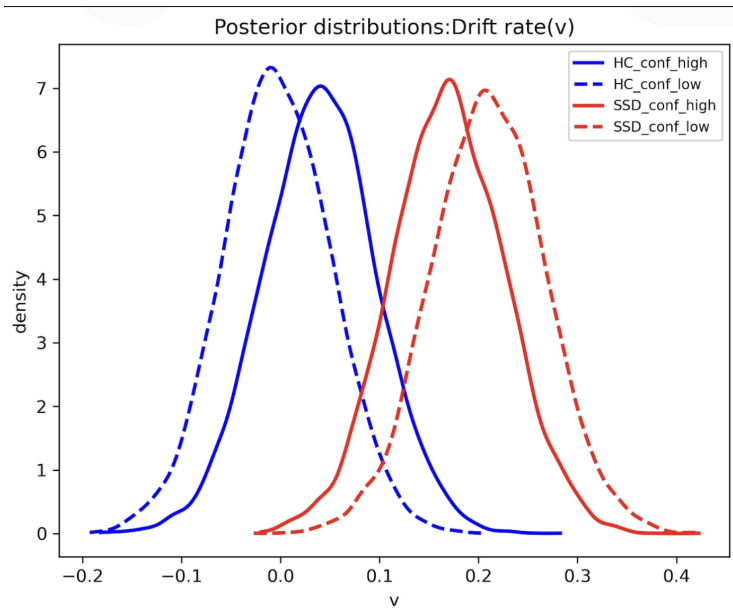


Fig 5. Posterior distributions: v

In HC participants, high previous confidence was associated with a higher drift rate, whereas the opposite pattern is observed in SSD. The evidence for this however was limited. The posterior mean of the interaction was 0.084, with a 95% credible interval of $[-0.136, 0.302]$ and a posterior probability of 0.77 that the interaction was positive. Because the credible intervals included zero, the data does not provide strong evidence that the effect of previous confidence on drift rate reliably differed between groups. The observed pattern however should be considered non-confirmative. Additional analyses were conducted to investigate the apparent differences in drift rate between the conditions, these can be found in the Appendix.

3.1.b. Boundary separation:

Boundary separation was examined to determine whether previous confidence influenced the amount of evidence required before participants committed to a subsequent response. The posterior estimates were higher in SSD than HC across both previous confidence conditions.

Group	Previous confidence	Posterior mean (a)
HC	High	2.229
HC	Low	2.333
SSD	High	2.484
SSD	Low	2.624

Table 3. Previous confidence and posterior means by group

The pattern indicated in the Table shows that SSD participants generally needed more accumulated evidence before committing a decision than HC participants. Because the staircase adjusted stimulus difficulty to maintain comparable accuracy between groups, the higher boundary separation is not explained by differences in task difficulty. This higher boundary separation shows a difference in decision strategy, SSD participants adopt a more cautious response threshold.

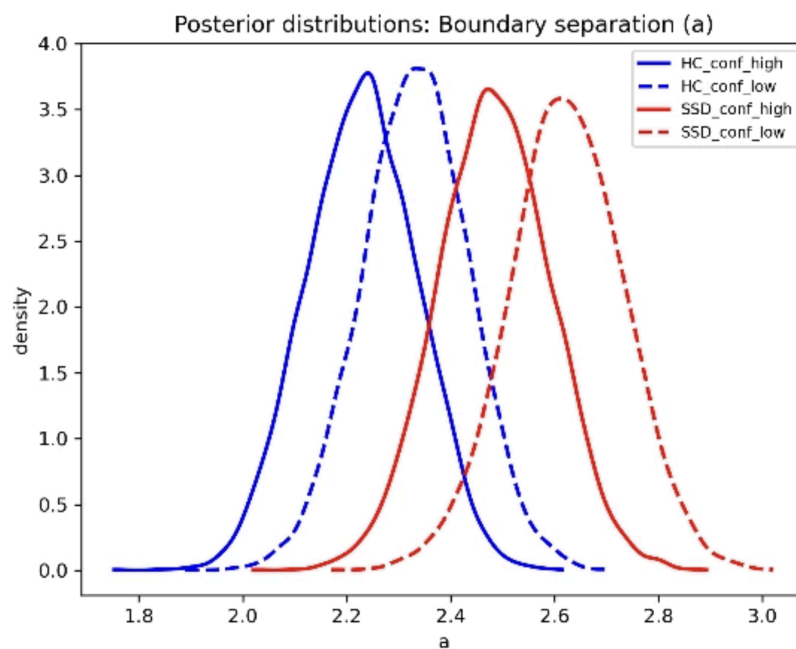


Fig 6. Posterior distributions: a

Previous confidence also showed a relationship with boundary separation within both groups, with lower previous confidence associated with higher estimated thresholds. This suggests that uncertainty about the preceding decision may carry forward to the next decision, leading participants to require more evidence before committing to a decision. The Stage 1 model therefore provides evidence that information from the previous confidence judgement can influence the subsequent decision process.

3.1.c. Non-decision time

A separate model allowed non-decision time to vary between groups to test whether group differences could be explained by processes outside evidence accumulation. Allowing ‘t’ to vary by group did not improve model fit relative to Model 4. The estimated mean non-decision times were also highly similar between HC and SSD:

Group	Mean (s)	SD
HC	1.348	0.098
SSD	1.345	0.105

Table 4. Non-decision times - means and standard deviation by group

There was therefore little evidence that the observed group differences were driven by differences in the non-decision component of the task. Given that Stage 1 reaction time was measured from stimulus onset to movement onset, this result provides limited evidence that

differences in perceptual encoding or processes preceding decision process account for observed group patterns.

3.1.d. Parameter identifiability:

Because drift rate and boundary separation may trade off in HDDM approaches when explaining reaction time distributions, posterior correlations between parameters were examined.

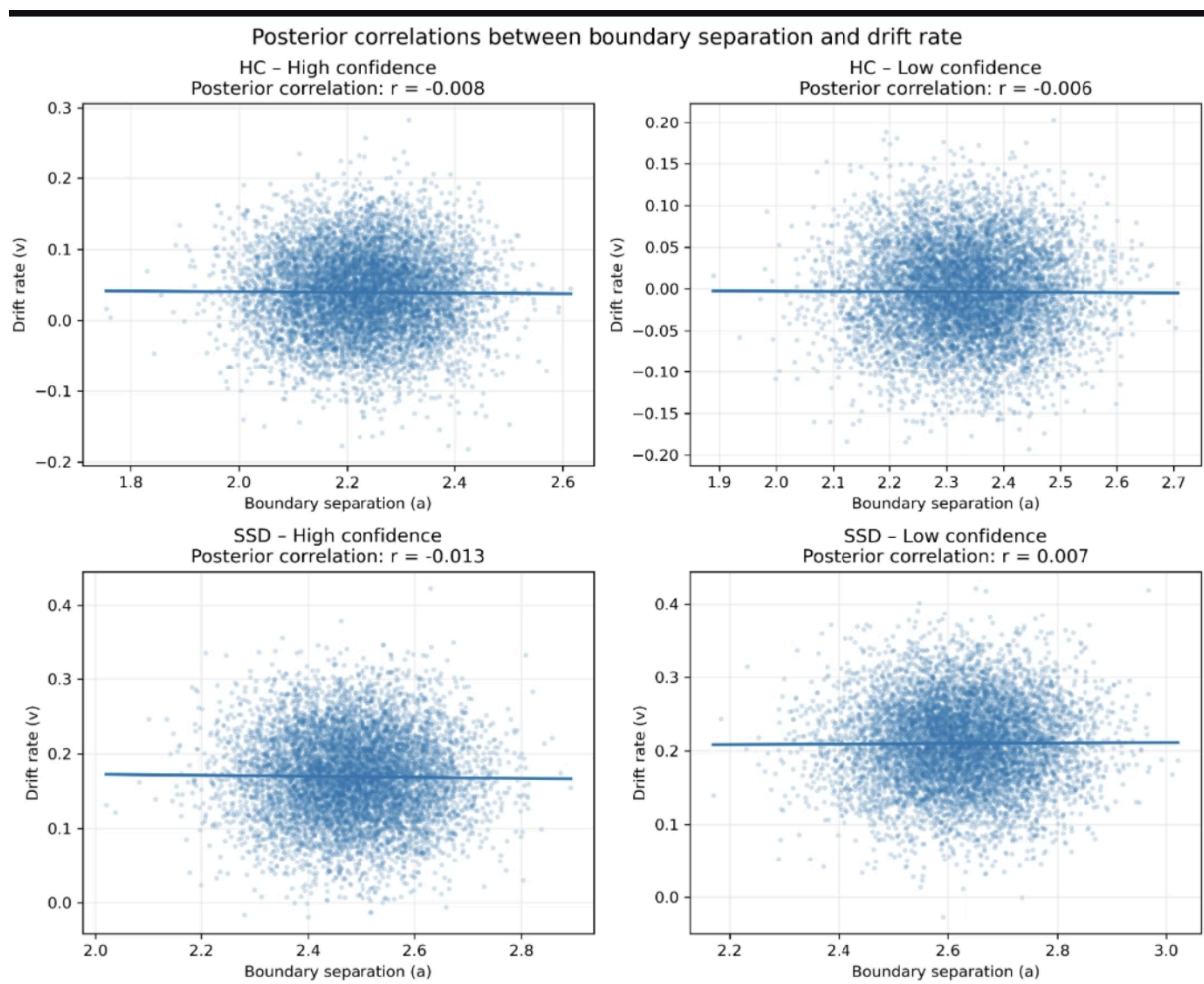


Fig 7. Posterior correlations between a and v

In this scatter plot, each dot shows one MCMC posterior sample from the HDDM Model 4, and the cloud represents the joint posterior distribution of the two parameters, ' v ' and ' a '.

The line is a linear regression line fitted to the posterior samples, it being horizontal on all

four conditions shows that the relationship between ‘a’ and ‘v’ is nearly zero. This is also what the ‘r’ , or Pearson correlation coefficient values indicate. These correlations suggest that the observed drift differences are unlikely to be caused solely by parameter trade-offs, so the group-specific drift patterns represent differences in the latent evidence accumulation process.

These findings are broadly consistent with the previous work of Faivre et al. (2023), in which previous confidence was considered in relation to subsequent decision formation and no strong overall group difference in task performance was identified. The present analysis extends this approach by separating observed choice and reaction-time behaviour into latent decision components. The absence of a substantial difference in non-decision time suggests that the observed differences are unlikely to be explained by slower processes occurring before or outside evidence accumulation.

Stage 1 therefore indicates that the decision process preceding confidence formation was not identical between the two groups. The boundary separation estimate was higher for SSD, indicating a more cautious response threshold. The HDDM model establishes whether previous confidence is associated with decision formation in the next trial. In this stage however, confidence is an observed variable, it does not explain where in the decision-evaluation cycle this confidence is generated. This question is addressed in Stage 2.

3.2 Stage 2 : 2DSD

Stage 2 tested whether evidence continues to accumulate after the initial perceptual decision and whether this post-decisional accumulation of evidence differs between HC and SSD. The 2DSD model characterises confidence accumulation as the accumulation of evidence post decision, do the post decisional drift rate corresponds to the confidence associated with that

decision. The model estimated the latent evidence state immediately following the decision (θ), the rate of post-decisional evidence accumulation (δ) and the amount of noise between estimated confidence and reported confidence (σ_{noise}).

The model estimated a positive initial evidence state, with $\theta = 0.424$ (89% ETI = [0.390, 0.460]), indicating that participants accumulated evidence after the perceptual decision was made. The post-decisional accumulation parameter was positive for both groups, $\delta_{HC} = 0.245$ (89% ETI = [0.220, 0.270]) and the $\delta_{SSD} = 0.198$ (89% ETI = [0.170, 0.220]). As both posterior intervals excluded zero, the results provide evidence that evidence accumulation continued beyond the initial decision in both groups.

	mean	sd
theta	0.424	0.021
δ_{HC}	0.245	0.0175
δ_{SSD}	0.198	0.0164
confidence_noise	0.0928	0.0024

Table 5. Mean and standard deviation values of δ by group

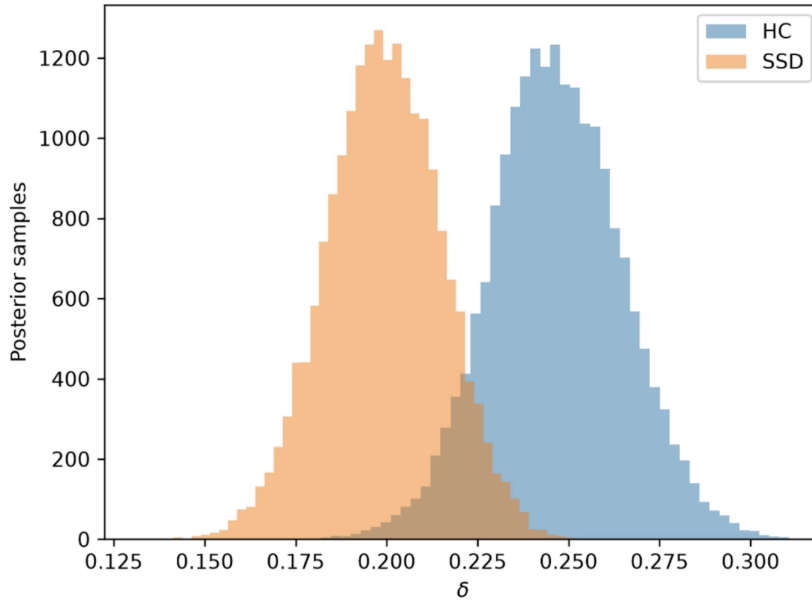


Fig 7. Posterior distribution for δ

The estimated posterior for HC is 0.245, while that for SSD is 0.198, which is nearly half that of HC. The estimated post decisional drift rate was numerically larger in HC than in the SSD group, suggesting that healthy participants accumulated evidence slightly more rapidly during the post-decisional period. Both groups had continued evidence accumulation during the post-decisional interval (from movement onset to click).

The observation model estimated a confidence noise parameter of $\sigma_{conf} = 0.0928$ (89% ETI = [0.089, 0.097]). Given that confidence ratings were modelled on a bounded scale between 0 and 1, this value indicated that reported confidence closely reflected the latent post decisional estimated confidence, with relatively low variability attributable to observation noise.

Sampling diagnostics supported the interpretation of the posterior estimates. All estimated parameters had $\hat{R}$ values of approximately 1.01, indicating good convergence of the four Markov chains and convergence to a common posterior distribution. Effective sample sizes (ESS) were also adequate (bulk ESS = 473-840; tail ESS = 1021-1773), suggesting that the posterior estimates were stable and interpretable.

Overall, Stage 2 provides evidence for continued evidence accumulation after the initial perceptual decision in both groups, with a lower estimated post-decisional drift rate in SSD than HC. With regard to the central research question, this suggests that confidence formation is associated with altered post-decisional evidence accumulation in SSD. This provides a process level explanation that could not be obtained in the Stage 1 HDDM alone, decomposing the decision process into initial perceptual decision and subsequent post decisional evidence accumulation or confidence formation was necessary.

4. Discussion

The present thesis investigated whether differences in confidence in schizophrenia-spectrum disorder (SSD) can be understood as a general impairment in metacognition, or whether they instead arise from alterations in specific computational processes involved in decision formation and confidence evaluation. Across two complementary modelling stages, the findings provide preliminary evidence for the latter interpretation. Stage 1 identified greater boundary separation in SSD than in healthy controls (HC), indicating greater response caution during perceptual decision formation. Stage 2 subsequently identified a lower post-decisional evidence accumulation rate (δ) in SSD, with the estimated rate approximately half that observed in HC. These findings suggest that confidence-related differences in SSD may not be adequately characterised by a single global measure of metacognitive efficiency. Instead, the decision-evaluation cycle may contain separable computational processes that are altered in different ways in SSD: individuals with SSD appeared more cautious in reaching the initial perceptual decision, but subsequently accumulated information for confidence at a slower rate.

This interpretation addresses an important tension within the literature on metacognition in schizophrenia. A substantial body of research (Moritz et al, 2006; Koren et al , 2004) has

reported impaired confidence calibration in schizophrenia, often interpreted as evidence for a metacognitive deficit. However, more recent work has highlighted the extent to which the apparent deficits depend on first-order task performance. The original study from which the dataset was considered in this thesis : the Bayesian meta-analysis by Faivre et al. (2021) found that the magnitude of reported metacognitive deficits was reduced when performance was equated between groups, with the evidence for a specific metacognitive impairment becoming inconclusive in performance-controlled studies. Similarly, Rouy et al. (2023) found no evidence for a specific metacognitive deficit across visual detection, familiarity and recollection tasks when performance was controlled, and reported preserved metacognitive efficiency in SSD. These findings argue against an account in which schizophrenia is characterised by a unitary failure to know whether one's decisions are correct.

The present findings extend this perspective by suggesting that the absence of a global metacognitive impairment does not necessarily imply that the computational processes generating confidence are unchanged.

4.1 Altered decision formation in SSD

The Stage 1 HDDM analysis provided evidence for greater boundary separation in SSD. In the drift diffusion decision-making framework, boundary separation reflects the amount of accumulated evidence required before committing to a response. The higher boundary separation estimated in SSD therefore suggests that participants with SSD adopted a more cautious response strategy, requiring more evidence before reaching the perceptual decision threshold.

This finding shifts the interpretation of altered confidence away from confidence reporting. A participant's confidence report is generated after, or in relation to, an underlying decision, and

therefore the properties of that decision can constrain the information later available for confidence evaluation. Greater response caution could mean that individuals with SSD spend longer accumulating evidence before committing to a perceptual choice. The finding is consistent with previous evidence that decision-making abnormalities in schizophrenia are not reducible to a single impairment in metacognition (Faivre et al., 2020; Rouy et al., 2023). Earlier work (Horga et al., 2014) using evidence-accumulation models, for example, has reported increased boundary separation and reduced drift rate in schizophrenia in some decision-making paradigms, whereas Faivre et al. found no group differences in the parameters of an evidence-accumulation model applied to a perceptual confidence task.

This distinction is particularly important when interpreting the Stage 1 drift-rate estimates. Drift rate is conventionally interpreted as the average rate at which evidence supporting one response accumulates towards the decision boundary. In principle, a group difference in v could therefore be interpreted as a difference in the quality or efficiency of evidence processing. However, this interpretation is substantially constrained in the present dataset because the original experiment used an adaptive staircase procedure. Stimulus difficulty was adjusted according to participants' preceding performance, meaning that the distribution of stimulus information encountered by each participant was partly determined by their behavioural history.

This limitation is relevant because the present study was not designed to test whether perceptual evidence processing itself is impaired in SSD. Its principal purpose was to determine whether confidence could be decomposed into distinct decision and post-decision processes. The boundary-separation finding is more interpretable for the central question than the group difference in drift rate. Future experiments using fixed and independently manipulated stimulus difficulty would be better suited to determining whether the altered

decision formation observed here reflects differences in evidence quality, evidence processing efficiency, response caution, or interactions between these factors.

4.2 Slower post-decisional evidence accumulation in SSD

The Stage 2 analysis provides the clearest evidence for a difference in the confidence-generation process itself. The estimated post-decisional accumulation parameter, (δ), was positive in both groups, indicating that evidence continued to accumulate after the perceptual decision had been reached. This is consistent with the theoretical framework proposed by Pleskac and Busemeyer (2010), in which decision formation and confidence evaluation are related but temporally distinct stages. In this framework, information available after the initial choice can continue to modify the evidence supporting the selected response and thereby contribute to confidence. The lower δ in SSD suggests that additional information was incorporated into confidence more slowly after the perceptual decision had been reached.

The finding provides a potentially important mechanistic interpretation of confidence differences in SSD. If confidence is partly constructed through the continued evaluation of information after a choice, then a reduction in the rate of post-decisional accumulation could alter the extent to which subsequent evidence modifies an initial decision. This provides a possible explanation for why confidence abnormalities may not always appear as a global reduction in metacognitive efficiency. A person may retain the capacity to distinguish correct from incorrect decisions while using post-decisional information differently when constructing the strength of their confidence.

4.3 In relation with computational accounts of schizophrenia

This interpretation is particularly relevant to work demonstrating that confidence in schizophrenia is influenced by the information available during and after decision formation. Zheng et al. (2022) reported that individuals with schizophrenia relied more heavily on recent confidence history and less heavily on task-related evidence when evaluating confidence in a temporal-order memory task. Importantly, this greater dependence on confidence history occurred despite patients showing comparable discrimination of correct and incorrect decisions when first-order performance was controlled. Their findings therefore suggest that an apparently preserved capacity to distinguish correct from incorrect responses can coexist with an altered strategy for constructing confidence.

Zheng et al. demonstrated that the *sources of information* used to construct confidence can differ in schizophrenia, particularly through greater reliance on recent confidence history. The present analysis instead identifies a difference in the *temporal dynamics of evidence integration* after the decision. Together, these findings support the broader proposition that confidence in SSD should not be understood solely in terms of whether confidence is accurate, but also in terms of which information is used to construct it and when that information is incorporated. These findings may indicate that the calibration of confidence to performance can remain relatively intact while the computational route by which confidence is constructed differs between groups.

The present results extend this interpretation by identifying a candidate temporal mechanism underlying such differences. Rather than asking whether confidence is "impaired", the present study asks whether the processes generating confidence operate differently. The slower (δ) estimate suggests that post-decisional information may have a reduced or slower influence on confidence in SSD.

The findings can also be situated within broader computational accounts of abnormal inference in schizophrenia. Jardri et al. (2017), used a computational model of confidence in which sensory evidence and prior information were combined through ascending and descending inference loops. They found stronger ascending circular inference in schizophrenia, with this parameter associated with positive symptoms, while descending and ascending loops were related to different symptom dimensions. Their findings suggest that abnormal confidence can arise from the way information is weighted and integrated during inference rather than from a simple global inability to assess one's own performance.

The present results extend the literature from the question of *what information is weighted* towards the question of *when information is accumulated*. In Jardri et al., the critical issue was abnormal interaction between sensory evidence and prior information. In Zheng et al., the critical issue was greater reliance on recent confidence history during metamemory evaluation. In the present study, the critical finding is slower post-decisional evidence accumulation. These are different mechanisms, but they converge on a broader view of confidence as a dynamically constructed inference rather than a unitary metacognitive faculty.

This temporal interpretation is also consistent with the broader two-stage literature on confidence. Pleskac and Busemeyer's 2DSD model formalised the proposal that evidence accumulation can continue after the choice threshold has been reached, while subsequent work has shown that post-choice information can causally affect confidence. The present findings suggest that this post-decisional component may be particularly informative when investigating SSD, because the rate of this process differed between groups even though previous work has not consistently identified a global metacognitive deficit.

4.4 A revised interpretation of metacognition in SSD

The findings of present work suggest a possible dissociation between decision commitment and post-decisional evaluation. SSD participants required more evidence before committing to the initial perceptual decision, but once that decision had been made, the estimated rate at which further evidence contributed to confidence was lower. These processes could have different behavioural consequences. Greater boundary separation may produce more cautious or delayed commitment, whereas slower post-decisional accumulation may reduce the rate at which subsequent information modifies the subjective strength of a decision.

The study therefore advances knowledge by moving from a static conception of confidence towards a process-based account of confidence dynamics in SSD.

4.5 Methodological contribution

A further contribution concerns the treatment of confidence as a continuous behavioural variable. The original 2DSD framework was developed using discrete confidence categories, whereas the present analysis adapted the model to retain the continuous confidence reports available in the original dataset. This avoids imposing arbitrary confidence categories and preserves variation in the strength of reported confidence across trials.

This was valuable for the present research question because the aim was to characterise confidence as a dynamic process. Retaining continuous confidence makes it possible to model variation in the relationship between post-decisional accumulation and the eventual confidence report without first discarding within-category information.

The model provides one plausible computational account of the observed confidence distributions, but alternative models could potentially explain the same behavioural data. In particular, confidence may be influenced by decision-time information, residual evidence available at the decision boundary, memory of recent decisions, response heuristics, or other

post-decisional processes. The current results therefore support the utility of the framework but do not establish that post-decisional evidence accumulation is the unique mechanism responsible for the observed confidence reports.

4.6. Limitations and future directions

The relatively small sample remains a substantial limitation. The dataset contained 21 HC and 19 SSD participants, and this restricts the precision with which between-participant distributions and group-level differences in latent parameters can be estimated. Hierarchical modelling improves estimation efficiency but does not remove the fundamental limitation imposed by a small number of participants. The posterior estimates should therefore be regarded as preliminary evidence requiring replication in larger samples.

A second limitation is the absence of demographic and clinical covariates in the group-level computational comparisons. Because participants were analysed primarily according to diagnostic group, the present data cannot establish whether the observed computational differences are specifically related to schizophrenia-spectrum pathology or vary with factors such as symptom severity, illness duration, medication exposure or demographic characteristics. Larger samples would allow computational parameters to be modelled continuously against clinical dimensions rather than treating diagnosis as the sole explanatory variable.

The adaptive staircase represents a particularly important methodological limitation. Its primary purpose was to control task difficulty and maintain comparable subjective accuracy between groups, which is advantageous for examining metacognition. However, because stimulus difficulty was adjusted according to participants' preceding performance, stimulus strength was not independent of behavioural history. This creates an interpretational problem

for drift rate because v is estimated from the relationship between stimulus information and response behaviour. Differences in v could reflect genuine differences in evidence processing, differences in the stimulus distributions experienced by the two groups, or interactions between participant behaviour and the staircase algorithm.

Finally, the Stage 2 model represents one computational account of confidence rather than an experimentally established decomposition of the confidence process. The finding of a lower δ therefore cannot establish that post-decisional evidence accumulation is uniquely responsible for differences in confidence in SSD. Alternative models may attribute the same behavioural variation to differences in pre-decisional evidence, decision-time information, confidence criteria, memory of previous decisions, or other mechanisms. This makes formal comparison between competing confidence models an important next step.

Future research should therefore move from identifying candidate computational differences towards testing their specificity and behavioural consequences. Larger samples should be used to improve the precision of group-level estimates and enable computational parameters to be related to continuous symptom dimensions rather than diagnostic categories alone. Simulation-based parameter recovery should also be conducted before data collection to determine whether the planned task and sample size provide sufficient information to distinguish the parameters of interest. Removing the adaptive staircase, or incorporating independently controlled levels of evidence strength, would make it possible to determine whether the increased response caution and any differences in drift rate persist when the information available to the two groups is experimentally matched rather than dynamically adjusted.

Finally, competing computational accounts should be compared. Models in which confidence depends primarily on evidence available at the time of the decision should be compared with

models incorporating post-decisional accumulation, confidence-history effects, or other sources of information. A future model-comparison framework could test whether the slower post-decisional accumulation identified here provides the best explanation of confidence dynamics, or whether it reflects a broader alteration in the weighting of current and historical information.

4.7 Conclusion

The present thesis provides preliminary evidence that confidence in SSD is better understood as the outcome of multiple computational processes than as a unitary metacognitive capacity. Individuals with SSD showed greater response caution during perceptual decision formation, reflected in increased boundary separation, and a substantially lower estimated rate of post-decisional evidence accumulation than HC. These findings suggest that the decision-evaluation cycle is not uniformly altered in SSD: the initial decision was characterised by greater caution, whereas the subsequent incorporation of information into confidence was slower.

The principal advance of the thesis is the demonstration that confidence can be decomposed into separable computational stages, and that these stages may be altered differently in SSD. The reduced post-decisional accumulation rate is particularly important because it provides a quantifiable hypothesis for future experimental and clinical research. Whether this mechanism relates to belief updating, reconsideration of decisions, behavioural persistence or treatment engagement remains unknown, but it can now be tested directly.

These findings should be regarded as preliminary. The small sample, adaptive staircase, uncertainty surrounding the interpretation of drift rate, and assumptions of the continuous-confidence 2DSD model constrain the strength of the conclusions. Nevertheless, the study provides a framework for moving beyond the question of whether people with SSD

have "impaired metacognition" towards a more precise question: which computational processes determine how evidence becomes a decision, and how that decision subsequently becomes a confidence judgement? Such work would establish whether the computational decomposition developed here can move beyond describing differences in laboratory decision-making towards explaining how altered evidence evaluation may contribute to the broader behavioural difficulties associated with SSD.

5. References

- Adams, R. A., Stephan, K. E., Brown, H. R., Frith, C. D., & Friston, K. J. (2013). The computational anatomy of psychosis. *Frontiers in Psychiatry*, 4(47).
<https://doi.org/10.3389/fpsy.2013.00047>
- Balzan, R. P., Woodward, T. S., Delfabbro, P., & Moritz, S. (2016). Overconfidence across the psychosis continuum: A calibration approach. *Cognitive Neuropsychiatry*, 21(6), 510–524. <https://doi.org/10.1080/13546805.2016.1240072>
- Bhatt, R., Laws, K. R., & McKenna, P. J. (2010). False memory in schizophrenia patients with and without delusions. *Psychiatry Research*, 178(2), 260–265.
<https://doi.org/10.1016/j.psychres.2009.02.006>
- Desender, K., Boldt, A., Verguts, T., & Donner, T. H. (2019). Confidence predicts speed-accuracy tradeoff for subsequent decisions. *eLife*, 8.
<https://doi.org/10.7554/elife.43499>
- Faivre, N., Roger, M., Pereira, M., de Gardelle, V., Vergnaud, J.-C., Passerieux, C., & Roux, P. (2021). Confidence in visual motion discrimination is preserved in individuals with

- schizophrenia. *Journal of Psychiatry and Neuroscience*, 46(1), E65–E73.
<https://doi.org/10.1503/jpn.200022>
- Fleming, S. M., & Lau, H. C. (2014). How to measure metacognition. *Frontiers in Human Neuroscience*, 8(1). <https://doi.org/10.3389/fnhum.2014.00443>
- Grant, P. M. P. (2012). Randomized trial to evaluate the efficacy of cognitive therapy for low-functioning patients with schizophrenia. *Archives of General Psychiatry*, 69(2), 121–127. <https://doi.org/10.1001/archgenpsychiatry.2011.129>
- Gurrera, R. J., McCarley, R. W., & Salisbury, D. (2014). Cognitive task performance and symptoms contribute to personality abnormalities in first hospitalized schizophrenia. *Journal of Psychiatric Research*, 55, 68–76.
<https://doi.org/10.1016/j.jpsychires.2014.03.022>
- Horga, G., Schatz, K. C., Abi-Dargham, A., & Peterson, B. S. (2014). Deficits in predictive coding underlie hallucinations in schizophrenia. *Journal of Neuroscience*, 34(24), 8072–8082. <https://doi.org/10.1523/jneurosci.0200-14.2014>
- Hoven, M., Lebreton, M., Engelmann, J. B., Denys, D., Luigjes, J., & van Holst, R. J. (2019). Abnormalities of confidence in psychiatry: An overview and future perspectives. *Translational Psychiatry*, 9(1). <https://doi.org/10.1038/s41398-019-0602-7>
- Kiani, R., & Shadlen, M. N. (2009). Representation of confidence associated with a decision by neurons in the parietal cortex. *Science*, 324(5928), 759–764.
<https://doi.org/10.1126/science.1169405>
- Kircher, T. T. J., Koch, K., Stottmeister, F., & Durst, V. (2007). Metacognition and reflexivity in patients with schizophrenia. *Psychopathology*, 40(4), 254–260.
<https://doi.org/10.1159/000101730>
- Koren, D., Seidman, L. J., Poyurovsky, M., Goldsmith, M., Viksman, P., Zichel, S., & Klein, E. (2004). The neuropsychological basis of insight in first-episode schizophrenia: A

- pilot metacognitive study. *Schizophrenia Research*, 70(2–3), 195–202.
<https://doi.org/10.1016/j.schres.2004.02.004>
- Maniscalco, B., & Lau, H. (2012). A signal detection theoretic approach for estimating metacognitive sensitivity from confidence ratings. *Consciousness and Cognition*, 21(1), 422–430. <https://doi.org/10.1016/j.concog.2011.09.021>
- Martiadis, V., Pessina, E., Raffone, F., Inziato, V., Martini, A., & Scognamiglio, P. (2023). Metacognition in schizophrenia: A practical overview of psychometric metacognition assessment tools for researchers and clinicians. *Frontiers in Psychiatry*, 14.
<https://doi.org/10.3389/fpsy.2023.1155321>
- Moritz, S., Balzan, R. P., Bohn, F., Veckenstedt, R., Kolbeck, K., Bierbrodt, J., & Dietrichkeit, M. (2016). Subjective versus objective cognition: Evidence for poor metacognitive monitoring in schizophrenia. *Schizophrenia Research*, 178(1–3), 74–79. <https://doi.org/10.1016/j.schres.2016.08.021>
- Pleskac, T. J., & Busemeyer, J. R. (2010). Two-stage dynamic signal detection: A theory of choice, decision time, and confidence. *Psychological Review*, 117(3), 864–901.
<https://doi.org/10.1037/a0019737>
- Renaud, J., Duverne, S., Litvinova, A. S., & Denève, S. (2017). Experimental evidence for circular inference in schizophrenia. *Nature Communications*, 8(1), 14218.
<https://doi.org/10.1038/ncomms14218>
- Rouy, M., Pereira, M., Saliou, P., Sanchez, R., el Mardi, W., Sebban, H., Baqué, E., Dezier, C., Porte, P., Micaux, J., de Gardelle, V., Mamassian, P., Moulin, C. J. A., Dondé, C., Roux, P., & Faivre, N. (2023). Confidence in visual detection, familiarity and recollection judgments is preserved in schizophrenia spectrum disorder. *Schizophrenia*, 9(1). <https://doi.org/10.1038/s41537-023-00387-4>

- Rouy, M., Saliou, P., Nalborczyk, L., Pereira, M., Roux, P., & Faivre, N. (2021). Systematic review and meta-analysis of metacognitive abilities in individuals with schizophrenia spectrum disorders. *Neuroscience and Biobehavioral Reviews*, 126, 329–337.
<https://doi.org/10.1016/j.neubiorev.2021.03.017>
- Spiegelhalter, D. J., Best, N. G., Carlin, B. P., & van der Linde, A. (2002). Bayesian measures of model complexity and fit. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(4), 583–639. <https://doi.org/10.1111/1467-9868.00353>
- Sterzer, P., Adams, R. A., Fletcher, P., Frith, C., Lawrie, S. M., Muckli, L., Petrovic, P., Uhlhaas, P., Voss, M., & Corlett, P. R. (2018). The predictive coding account of psychosis. *Biological Psychiatry*, 84(9), 634–643.
<https://doi.org/10.1016/j.biopsych.2018.05.015>
- Warman, D. M. (2008). Reasoning and delusion proneness: Confidence in decisions. *The Journal of Nervous and Mental Disease*, 196(1), 9–15.
<https://doi.org/10.1097/NMD.0b013e3181601141>
- Weschler. (2008). Weschsler adult intelligence scale: 4Th edition (WISC-IV) | pearson assessments US. In *Pearsonassessments.com*.
https://www.pearsonassessments.com/en-us/Store/Professional-Assessments/Cognition-%26-Neuro/Wechsler-Adult-Intelligence-Scale-%7C-Fourth-Edition/p/100000392?utm_source=chatgpt.com
- Wiecki, T. V., Sofer, I., & Frank, M. J. (2013). HDDM: Hierarchical bayesian estimation of the drift-diffusion model in Python. *Frontiers in Neuroinformatics*, 7.
<https://doi.org/10.3389/fninf.2013.00014>
- Zheng, Y., Wang, L., Gerlofs, D. J., Duan, W., Wang, X., Yin, J., Yan, C., Allé, M. C., Berna, F., Wang, J., Tang, Y., & Kwok, S. C. (2021). Atypical meta-memory evaluation

strategy in schizophrenia patients. *Schizophrenia Research. Cognition*, 27, 100220.

<https://doi.org/10.1016/j.scog.2021.100220>

6. Appendix

Appendix A

Data preprocessing steps:

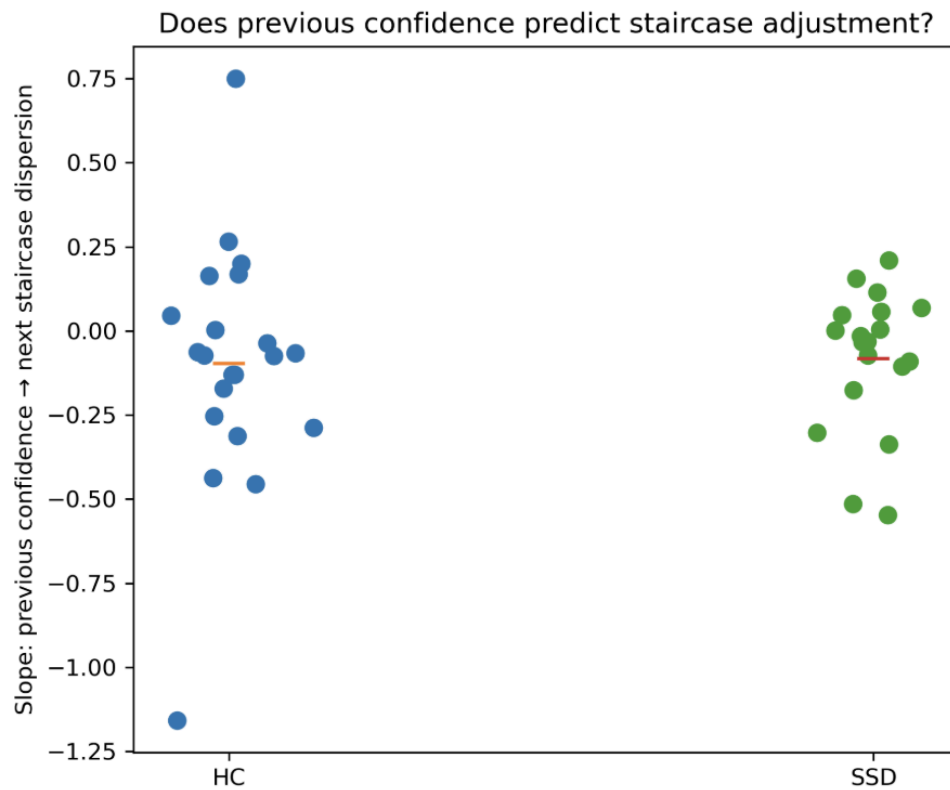
The preprocessing step was implemented using python (pandas, NumPy, and SciPy). Participant-specific fences were calculated separately for decision RT and tau using a log-transformed median absolute deviation (MAD) with $k=2.5$. A minimum of 10 valid observations was required to calculate a participant-specific fence; otherwise, the predefined hard limits were applied. All exclusion criteria were stored as binary flags, with participant-level exclusion counts saved to `exclusion_log.csv` to maintain an audit trail.

Table A1: Participant wise number of trials retained after preprocessing.

participant_id	group	Total trials in dataset	Trials excluded	Trials retained	Percentage retained
1	HC	300	15	285	95
2	HC	300	19	281	93.67
4	HC	300	3	297	99
5	HC	300	4	296	98.67
8	HC	300	4	296	98.67
9	HC	300	25	275	91.67
12	HC	300	1	299	99.67

16	HC	300	14	286	95.33
18	HC	300	7	293	97.67
19	HC	300	10	290	96.67
23	HC	300	4	296	98.67
25	HC	300	15	285	95
26	HC	300	26	274	91.33
29	HC	300	1	299	99.67
30	HC	300	15	285	95
31	HC	300	38	262	87.33
32	HC	300	1	299	99.67
35	HC	300	4	296	98.67
36	HC	300	12	288	96
37	HC	300	12	288	96
38	HC	300	13	287	95.67
0	SSD	300	32	268	89.33
3	SSD	300	26	274	91.33
6	SSD	300	17	283	94.33
7	SSD	300	19	281	93.67
10	SSD	300	10	290	96.67
11	SSD	300	22	278	92.67
13	SSD	300	19	281	93.67
14	SSD	240	24	216	90
15	SSD	300	7	293	97.67
17	SSD	300	46	254	84.67
20	SSD	300	8	292	97.33
21	SSD	300	26	274	91.33
22	SSD	300	4	296	98.67
24	SSD	300	50	250	83.33
27	SSD	300	3	297	99
28	SSD	300	27	273	91
33	SSD	300	11	289	96.33
34	SSD	300	23	277	92.33
39	SSD	300	8	292	97.33

Figure A1:



Participant-level relationship between previous-trial confidence and subsequent staircase adjustment. Each point represents the estimated regression slope for an individual participant, quantifying the relationship between confidence reported on trial $t-1$ and the change in staircase dispersion applied to the subsequent trial.

Table A2: Stage 1 Model parameters as approximated by HDDM

Model	parameter	mean	std
Model 1	a	2.428	0.486
	v_HC	0.008	
	v_SSD	0.188	
	v		0.226

Model 2	a_conf_high	2.365	
	a_conf_low	2.471	
	a		0.494
	v_HC	0.01	
	v_SSD	0.190	
	v		0.228
Model 3	a(HC_conf_high)	2.233	
	a(HC_conf_low)	2.328	
	a(SSD_conf_high)	2.483	
	a(SSD_conf_low)	2.613	
	a		0.477
	v_HC	0.009	
	v_SSD	0.190	

	v		0.228
Model 4	a_HC_conf_high	2.228	
	a_HC_conf_low	2.332	
	a_SSD_conf_high	2.484	
	a_SSD_conf_low	2.623	
	a		0.477
	v_HC_conf_high	0.039	
	v_HC_conf_low	-0.003	
	v_SSD_conf_high	0.169	
	v_SSD_conf_low	0.209	
	v		0.236

These were the latent decision parameters estimated by each of the HDDM models run based on how drift rate ‘v’ and boundary separation ‘a’ differed with respect to group and confidence on the previous trial.

Figure A2: Posterior distributions of ‘v’ and ‘a’ in Model 4 of Stage 1.

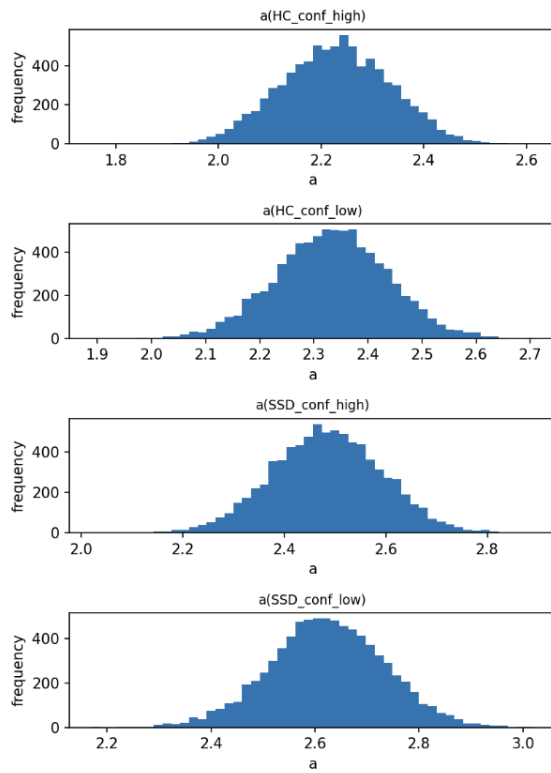
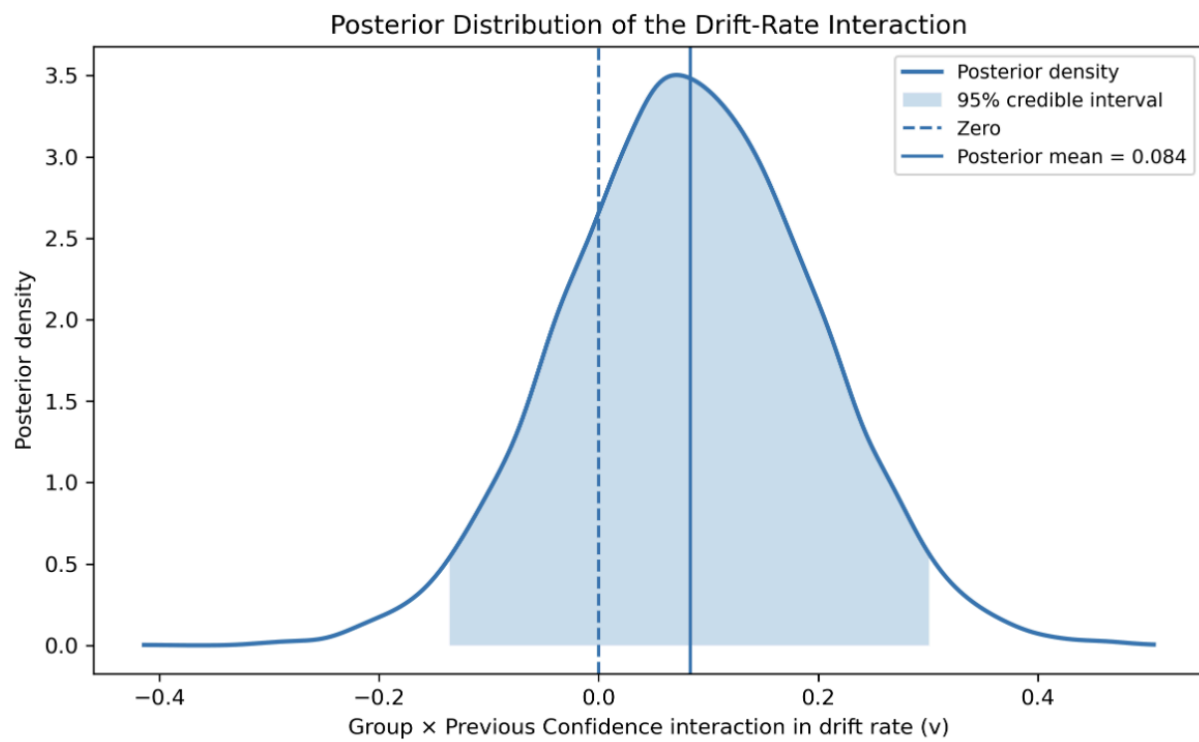


Figure A shows the posterior distributions of the HDDM boundary separation parameter (a) for healthy control (HC) participants and participants with schizophrenia spectrum disorder (SSD), separately according to whether the preceding trial was associated with high or low confidence. Each histogram represents samples drawn from the posterior distribution of the corresponding group $\times$ previous-confidence parameter.

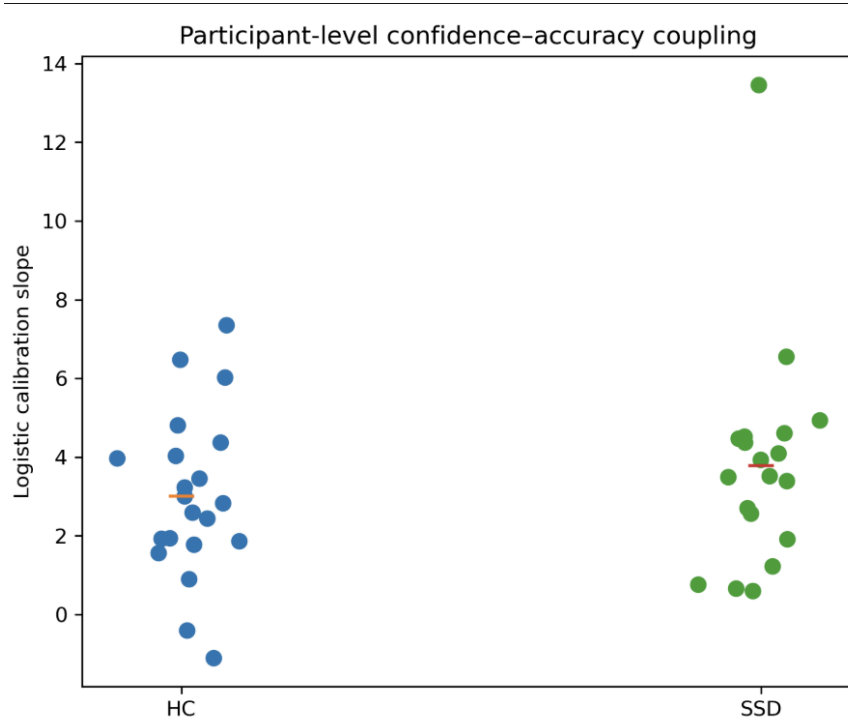
Figure A3: Posterior Distribution of the drift rate interaction



Supplementary analyses were conducted to verify the apparent opposite interaction of group and previous confidence in SSD versus HC groups, in Fig. 5 Posterior distributions of v , in Stage 1, Model 4 of the HDDM.

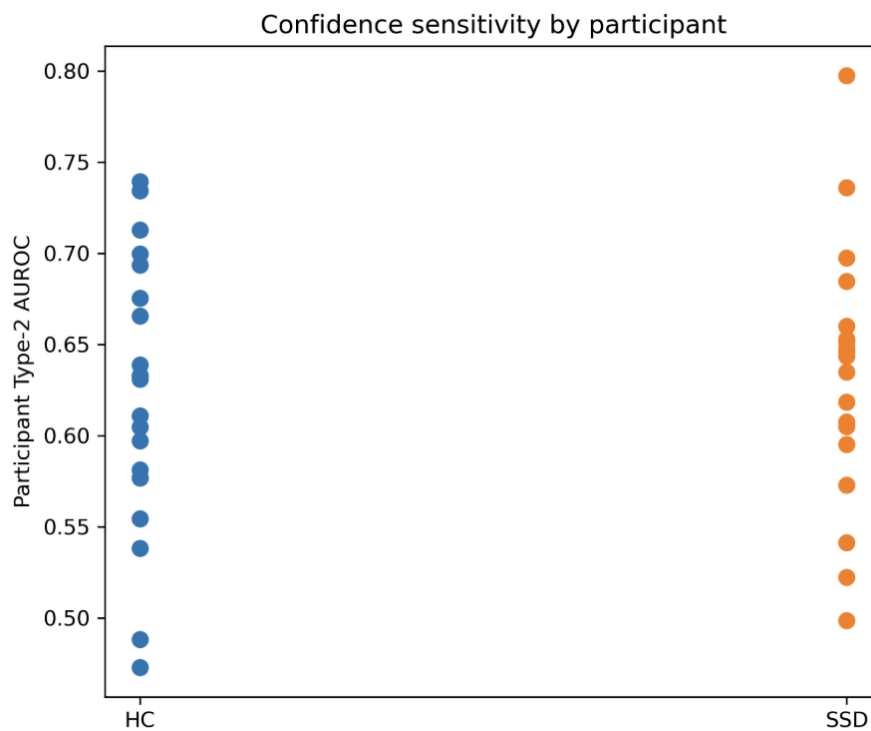
Appendix B

Figure B1: Participant-level confidence-accuracy coupling measured using Logistic calibration slope



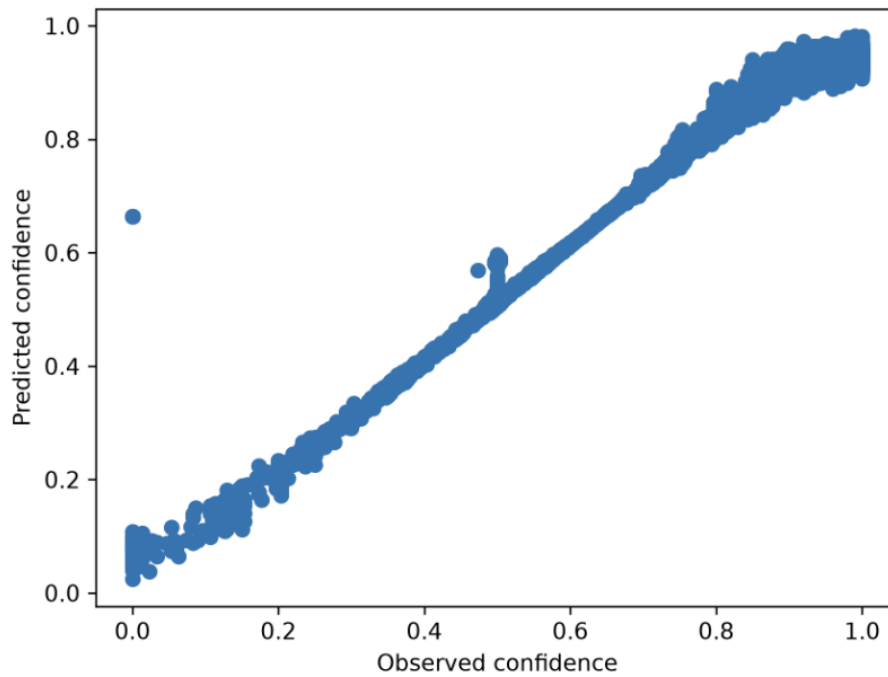
Participant-level confidence–accuracy coupling. Each point represents the logistic calibration slope estimated separately for an individual participant, quantifying the extent to which trial-by-trial confidence predicted decision accuracy. Higher positive slopes indicate a stronger positive relationship between confidence and accuracy, such that higher confidence was more strongly associated with correct responses. Values close to zero indicate weaker confidence–accuracy coupling, whereas negative values indicate an inverse relationship. Horizontal markers indicate the group-level mean.

Figure B2: Confidence sensitivity by participant measured using Type 2 AUROC



Participant-level confidence sensitivity measured using Type-2 AUROC (AUROC2). Each point represents an individual participant's Type-2 area under the receiver operating characteristic curve, quantifying the extent to which confidence discriminated between correct and incorrect decisions. Values of 0.50 indicate chance-level confidence sensitivity, whereas values above 0.50 indicate that higher confidence was preferentially associated with correct responses.

Figure B3: Plot of Predicted confidence by the 2DSD model and observed confidence values



The figure compares trial-level observed confidence ratings with the corresponding confidence values predicted by the Stage 2 post-decisional evidence accumulation model. Each point represents an observation, with the horizontal axis showing observed confidence and the vertical axis showing the model-predicted confidence.